6.5 Analysis

Students may be given a set of experimental results to analyse.

Candidates will be assessed on their ability to:

Analyse data	• perform calculations, using the correct number of significant figures • plot results on a graph using an appropriate scale and units – the graph could be logarithmic in nature • use the correct units throughout • comment on the trend/pattern obtained • determine the relationship between two variables or determine a constant with the aid of the graph, e.g. by determining the gradient using a large triangle • use the terms precision, accuracy and sensitivity appropriately • suggest realistic modifications to reduce errors • suggest realistic modifications to improve the experiment • discuss uncertainties qualitatively and quantitatively • compound percentage uncertainties correctly • determine the percentage uncertainty in measurements for a single reading using **half** the resolution of the instrument **and** from multiple readings using the **half** range.